FUJITSU

rApp Software Development Kit

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User Guide

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Document Change Notice

Location of Change	Description of Change
Throughout	■ Initial release

Chapter 1
Overview

Python® Software Development Kit (SDK) is a Python-based software package that can be imported into any Non Real Time-RAN Intelligent Controller (NonRT-RIC) Application (rApp) code. This package provides an easy way of invoking R1 APIs towards the Fujitsu NonRT-RIC. With the Python SDK, the rApp developer can focus on the core of the rApp business logic without worrying about the details of interfacing with NonRT-RIC.

The rApp SDK provides ready-to-use functions that wrap the R1 APIs towards Service Management and Exposure (SME) and Data Management and Exposure (DME) microservices within the NonRT-RIC. It is integrated with keycloak for user authentication to perform token management, agnostic to rApp core functions.

The rApp SDK performs periodic heartbeat with NonRT-RIC to maintain the active state of the rApp with NonRT-RIC. It has a host of other rApp developer-friendly functions.

The Python SDK provides wrapper functions for the following group of APIs:

- R1-SME: This group of functions enables rApp to register/deregister with the NonRT-RIC, register/deregister, discover, and subscribe the services.
- R1-DME: This group of functions enables rApp to register/deregister data types and send or receive the data.
- rApp SDK:
  - Provides functions to fetch historical data based on the start and end time of the generated data.
  - Provides O1-related functions to fetch performance, fault, and configuration data.
  - Performs token generation for authentication and automatic token regeneration on expiry.
  - Performs the heartbeat function with NonRT-RIC to maintain the active state of the rApp with NonRT-RIC.

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Chapter 2
Install Python SDK Library

This procedure describes how to install the Python SDK library.

Step 1
Download Python SDK library from the Fujitsu *Partners* website, https://partners.fnc.fujitsu.com/selfservice using user credentials (Partners account required).

Step 2
Open terminal command prompt.

Step 3
Change the directory.

Command:
```
$ cd rapp-sdk
```

Step 4
Install the `pip` to upgrade setuptools wheel.

Command:
```
$ pip install --upgrade setuptools wheel
```

Step 5
Install the Python setup.

Command:
```
$ python setup.py install
```

Step Result:
The Python setup is installed.

Step 6
Clean up the repository.

Command:
```
$ cd
$ rm -r rapp-sdk/ -f
```

Step Result:
The repository is removed.

✓ **This task is complete.**

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Chapter 3
Service Management and Exposure Package

In this chapter:

3.1 Register rApp
3.2 Deregister rApp
3.3 Service Registration
3.4 Service Deregistration
3.5 Service Update
3.6 Service Discovery
3.7 Get Service with rApp ID
3.8 Subscribe Service
3.9 Unsubscribe Service

The following methods of the SDK invokes some of the APIs exposed by the SME.

Note: To use SME SDK functions, the SME client instance must be created from the SME client class provided by the rApp SDK by adding the following codes in the Python file.

Command:
```
from rappsdk.sme.sme_client import SMEClient
sme_client_instance = SMEClient(sme_url, username, password)
```

Table 1
Input Parameters

Key	Mandatory/Optional	Type	Description
`sme_url`	Mandatory	string	SME server URL
`username`	Mandatory	string	Username registered in keycloak
`password`	Mandatory	string	Password

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3.1
Register rApp

Method Name	`register_rapp(self, rAppDetails)`

Description

This method invokes SME API for registering an rApp with NonRT-RIC framework. This method takes an instance of `rAppRegister` class as an input and returns information registered with SME as a dictionary.

Table 2
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppDetails`	Mandatory	rAppRegister	`rAppRegister` instance containing the rApp details needed for registration with SME

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Table 3
Functional Outputs

<table>
<tr><th>Status</th><th>Type</th><th>Description</th></tr>
<tr><td>Successful</td><td>dictionary</td><td>Dictionary containing rApp information registered with SME in the following format:
<table>
<tr><th>Key</th><th>Type</th><th>Description</th></tr>
<tr><td>rAppName</td><td>string</td><td>Name of rApp provided in input</td></tr>
<tr><td>vendor</td><td>string</td><td>Vendor name</td></tr>
<tr><td>rAppVersion</td><td>string</td><td>Version of rApp</td></tr>
<tr><td>certificates</td><td>dictionary</td><td>Currently, dictionary is empty</td></tr>
<tr><td>serviceType</td><td>list[serviceTypeEnum]</td><td>List of enum types which indicates type of rApp</td></tr>
<tr><td>appType</td><td>string</td><td>Type of rApp</td></tr>
<tr><td>interfaceEndpoint</td><td>string</td><td>URL or endpoint of rApp</td></tr>
<tr><td>rAppId</td><td>string</td><td>Random string generated by SME to identify every unique rApp</td></tr>
<tr><td>listOfServices</td><td>list[string]</td><td>List of string which contains services registered by rApp. It is empty at the time of new registration.</td></tr>
<tr><td>heartbeatPeriod</td><td>string</td><td>Period in which rApp has to send health check requests to SME after every fixed time interval which is given in seconds.</td></tr>
</table></td></tr>
<tr><td>Successful</td><td>dictionary</td><td>
<table>
<tr><th>Key</th><th>Type</th><th>Description</th></tr>
<tr><td>rAppStatus</td><td>string</td><td>rAppStatus can be ACTIVE or INACTIVE. It is ACTIVE for new registration.</td></tr>
</table></td></tr>
<tr><td>Unsuccessful</td><td>Exceptions</td><td>Method raises an instance of exception if the operation is unsuccessful.</td></tr>
</table>

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3.2
Deregister rApp

Method Name	`deregister_rapp(self, rAppId)`

Description

This method invokes SME API for deregistering the given rApp from the NonRT-RIC framework. This method takes `rAppId` as input and returns information of the deregistered rApp as a dictionary.

Table 4
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Id of the rApp which must be deregistered from the NonRT-RIC framework

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Table 5
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing rApp details of the deregistered rApp in the following format:
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

Key	Type	Description
`rAppName`	string	Name of rApp provided in input
`vendor`	string	Vendor name
`rAppVersion`	string	Version of rApp
`certificates`	dictionary	Currently, dictionary is empty
`serviceType`	list[serviceTypeEnum]	List of enum types which indicates type of rApp
`appType`	string	Type of rApp
`interfaceEndpoint`	string	URL or endpoint of rApp
`rAppId`	string	Random string generated by SME to identify every unique rApp
`listOfServices`	list[string]	List of string which contains services registered by rApp
`heartbeatPeriod`	string	Period in which rApp has to send health check requests to SME after every fixed time interval which is given in `seconds`
`rAppStatus`	string	`INACTIVE`. It is `INACTIVE` for deregistered rApp

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3.3
Service Registration

Method Name	`register_service(self, serviceDetails)`

Description

This method invokes SME API for registering a service with the NonRT-RIC framework. This method takes an instance of `serviceDetails` class as an input and returns service details of the registered service as a dictionary.

Table 6
Input Parameters

Key	Mandatory/Optional	Type	Description
`serviceDetails`	Mandatory	serviceRegister	`serviceRegister` instance containing the service details needed for registration with SME

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Table 7
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing service details of the registered service in the following format:

Key	Type	Description
`rAppId`	string	Id of the rApp whose service must be registered
`serviceName`	string	Name of the service to be registered
`serviceVersion`	string	Version of the service
`accessMethod`	dictionary	Currently, dictionary is empty
`serviceType`	accessMethodsEnum	Indicates the access type for the given service
`communicationMechanism`	dictionary	Currently, dictionary is empty
`serviceEndpoint`	string	URL or endpoint of service
`serviceLimitations`	dictionary	Currently, dictionary is empty
`serviceCapabilities`	dictionary	Currently, dictionary is empty
`serviceId`	list[string]	Random string generated by SME to identify every unique service
`serviceStatus`	string	`serviceStatus` can be `ACTIVE` or `INACTIVE`. It is `ACTIVE` for new service registration.

Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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3.4
Service Deregistration

Method Name	`deregister_service(self, rAppId, serviceId)`

Description

This method invokes SME API for deregistering the given service associated with a given `rAppId` from the NonRT-RIC framework. This method takes `rAppId` and `serviceId` as input and returns the service details of the deregistered service as a dictionary.

Table 8
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Id of the rApp whose service must be deregistered from the NonRT-RIC framework
`serviceId`	Mandatory	string	Id of the service which must be deregistered

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Table 9
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing service details of the deregistered service in the following format:
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

Key	Type	Description
`rAppId`	string	Id of the rApp whose service must be deregistered
`serviceName`	string	Name of the service to be deregistered
`serviceVersion`	string	Version of the service
`accessMethod`	dictionary	Currently, dictionary is empty
`serviceType`	accessMethodsEnum	Indicates the access type for the given service
`communicationMechanism`	dictionary	Currently, dictionary is empty
`serviceEndpoint`	string	URL or endpoint of service
`serviceLimitations`	dictionary	Currently, dictionary is empty
`serviceCapabilities`	dictionary	Currently, dictionary is empty
`serviceId`	list[string]	Random string generated by SMEs to identify every unique service
`serviceStatus`	string	`serviceStatus` can be `ACTIVE` or `INACTIVE`. It is `ACTIVE` for deregistered services

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3.5
Service Update

Method Name	`update_service(self, rAppId, serviceId, serviceDelta)`

Description

This method invokes SME API for updating service details associated with an rApp. This method takes `rAppId`, `serviceId`, and `serviceDelta` as input and returns the updated service details of the given service as a dictionary.

Table 10
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Id of the rApp whose associated service details have to be updated
`serviceId`	Mandatory	string	Id of the service whose details have to be updated
`serviceDelta`	Mandatory	serviceUpdate	`serviceUpdate` instance containing the details of the service to be updated

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Table 11
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing updated service details of the given service in the following format:

Key	Type	Description
`rAppId`	string	Id of the rApp whose service must be updated.
`serviceName`	string	Name of the service to be updated.
`serviceVersion`	string	Version of the service
`accessMethod`	accessMethodsEnum	Indicates the access type for the given service
`communicationMechanism`	dictionary	Currently, dictionary is empty
`serviceEndpoint`	string	URL or endpoint of service
`serviceLimitations`	dictionary	Currently, dictionary is empty
`serviceCapabilities`	dictionary	Currently, dictionary is empty
`serviceId`	list[string]	Random string generated by SMEs to identify every unique service
`serviceStatus`	string	`serviceStatus` can be `ACTIVE` or `INACTIVE`. It is `ACTIVE` for existing active services.

Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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3.6
Service Discovery

Method Name	`discover_services(self, rAppId, serviceName)`

Description

This method invokes SME API for discovering the services registered by rApp (excluding the services registered by rApp that is doing the discovery) and provide the details. If a service name is given and the service is registered in that name, it only returns the details of that particular service. This method takes `rAppId` and `serviceName` (optional) as input and returns the `serviceDetails` as a list of the dictionary if the service name is given, else returns details of all services registered by rApp (excluding the services registered by rApp which is doing the discovery) as a list of dictionaries.

Table 12
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Id of the rApp who wants to discover the services
`serviceName`	Optional	string	Name of the service whose details have to be returned. If none, all the services registered with the NonRT-RIC are returned

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Table 13
Functional Outputs

Status	Type	Description
Successful	list[dictionary]	Service details of the given service as a list of dictionaries or details of all the services associated with the given rApp as a list of dictionaries given in the following format: **Key** / **Type** / **Description** `serviceName` / string / Name of the service `serviceVersion` / string / Version of the service `accessMethod` / accessMethodsEnum / Indicates the access type for the given service `communicationMechanism` / dictionary / Indicates the communication mechanism for the given service `serviceEndpoint` / string / URL or endpoint of service `serviceLimitations` / dictionary / Limitations of the service `serviceCapabilities` / dictionary / Capabilities of the service `serviceStatus` / string / `serviceStatus` can be `ACTIVE` or `INACTIVE`. It is `ACTIVE` for existing services
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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3.7
Get Service with rApp ID

Method Name	`getall_services(self, rAppId)`

Description

This method invokes SME API for discovering all the service details associated with an rApp. This method takes the `rAppId` as input. If `rAppId` is provided, it returns the service details of all the services associated with the given rApp as a list of dictionaries. If `rAppId` is not provided, the method returns a list of dictionaries listing all the services registered by all rApp.

Table 14
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Optional	string	Id of the rApp whose associated service details have to be discovered

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Table 15
Functional Outputs

Status	Type	Description
Successful	list[dictionary]	List of dictionaries containing services details of all the services associated with the given rApp or list of dictionaries containing service details of all the registered services in the following format:
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

Key	Type	Description
`rAppId`	string	Id of the rApp given in the input
`serviceName`	string	Name of the service
`serviceVersion`	string	Version of the service
`accessMethod`	accessMethodsEnum	Indicates the access type for the given service
`communicationMechanism`	dictionary	Indicates the communication type for the given service
`serviceEndpoint`	string	URL or endpoint of service
`serviceLimitations`	dictionary	Limitations of the service
`serviceCapabilities`	dictionary	Capabilities of the service
`serviceId`	list[string]	Random string generated by SME to identify every unique service
`serviceStatus`	string	`serviceStatus` can be `ACTIVE` or `INACTIVE`. It is `ACTIVE` for existing active services
`rAppName`	string	Name of the rApp

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3.8
Subscribe Service

Method Name	`subscribe_services(self, subscribe_service)`

Description

This method invokes SME API for subscribing to a service registered with the NonRT-RIC framework. This method takes an instance of the `subscribeRequest` class as input and returns the details of the subscription as a dictionary.

Table 16
Input Parameters

Key	Mandatory/Optional	Type	Description
`subscribe_service`	Optional	subscribeRequest	`subscribeRequest` instance containing the details of the service to be subscribed

Table 17
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing details of the subscription in the following format: **Key** / **Type** / **Description** `subscriptionId` / string / Unique Id generated by SME for each new subscription request `rAppId` / string / Id of the rApp, which has subscribed to the service `subscribedServices` / list[string] / List of the service names which have been subscribed by the rApp
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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3.9
Unsubscribe Service

Method Name	`unsubscribe_services(self, unsubscribe_service)`

Description

This method invokes SME API for unsubscribing from a subscribed service. This method takes an instance of `unSubscribeService` class having details of the service to be unsubscribed as an input and returns the `serviceId` of the unsubscribed service, `rAppId`, and list of remaining subscribed services as a dictionary.

Table 18
Input Parameters

Key	Null	Type	Description
`unsubscribe_service`	—	unSubscribeService	`unSubscribeService` instance containing the details of the service to be unsubscribed

Table 19
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing details in the following format: Key: `subscriptionId` — Type: string — Description: Subscription Id of the service which has been unsubscribed Key: `rAppId` — Type: string — Description: Id of the rApp, which has to unsubscribe from the service Key: `subscribedServices` — Type: list[string] — Description: List of the remaining service names which have been subscribed by the rApp
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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Chapter 4
Data Management and Exposure Package

In this chapter:

4.1 Register Data Type
4.2 Deregister Data Type
4.3 Subscribe Data Type Change
4.4 Unsubscribe Data Type Change
4.5 Discover Data Type
4.6 Subscribe Data
4.7 Pull Data
4.8 Push Data
4.9 Request Data
4.10 Get All Data Type IDs

The following SDK methods use or invokes the APIs exposed by DME.

Note: To use DME SDK functions, the DME client instance must be created from the DME client class provided by the rApp SDK by adding the following codes in the Python file.

Command:

```
from rappsdk.dme.dme_client import DMEClient
dme_client_instance = DMEClient(dme_url=, username=, password=)
```

Table 20
Input Parameters

Key	Null	Type	Description
`dme_url`	n/a	string	DME server URL
`username`	n/a	string	The registered username (keycloak credentials)
`password`	n/a	string	The password corresponds to the given username (keycloak credentials)

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4.1
Register Data Type

Method Name	`register_dataType(self, dataTypeDetails)`

Description

This method register a data type to the NonRT-RIC framework. Only a registered rApp with valid credentials can register a data type.

Table 21
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataTypeDetails`	Mandatory	dataTypeRegister	`dataTypeRegister` instance that contains the details of the data type for registration

Table 22
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary which contains details of the registered data type. dataTypeRegister Added fields are described in the following format. **Key** / **Type** / **Description** `accessDetails` / dictionary / Details of accessing data. The value depends on the `accessMethod` key, which is specified during registration. **RESTApi** **Key** / **Type** / **Description** `url` / string / URL endpoint to access data **MsgBus** **Key** / **Type** / **Description** `url` / string / Kafka host URL `topic` / string / Kafka topic name

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Table 22 (Cont.)
Functional Outputs

Status	Type	Description
Successful	Mandatory	Dictionary which contains details of the registered data type. dataTypeRegister Added fields are described in the following format. **Key** / **Type** / **Description** `accessDetails` / dictionary / Details of accessing data. The value depends on the `accessMethod` key, which is specified during registration. **Key** / **Type** / **Description** `url` / string / Websocket (WS) URL `topic` / string / WS topic name `token` / string / WS token `interfaceEndpoint` / string / Interface endpoint of the rApp, which is provided to SME when registering rApp
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.2
Deregister Data Type

Method Name	`deregister_dataType(self, dataTypeId)`

Description

This method deregister a data type from the NonRT-RIC framework.

Table 23
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataTypeId`	Mandatory	string	The `dataTypeId`, which is meant to be deregistered

Table 24
Functional Outputs

Status	Type	Description

Successful	list	The list contains a string message.
		
		Example:
		`[`
		`  "Deregistering data type: <rAppName> succeeded."`
		`]`

Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.3
Subscribe Data Type Change

Method Name	`subscribe_dataType_change(self, dataTypeId, rApp_id)`

Description

This method subscribe to a data type change from the NonRT-RIC framework to receive the list of active data types.

Table 25
Input Parameters

Key	Mandatory/Optional	Type	Description
`data_type_id`	Mandatory	string	The data type identifier should be subscribed for its availability notifications. This identifier could be both registered and not-registered
`rApp_id`	Mandatory	string	The identifier of the registered rApp

Table 26
Functional Outputs

Status	Type	Description
Successful	dictionary	Empty dictionary upon successful operation
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.4
Unsubscribe Data Type Change

Method Name	`unsubscribe_dataType_change(self, data_type_id, rApp_id)`

Description

This method unsubscribe from a data type change from the NonRT-RIC framework.

Table 27
Input Parameters

Key	Mandatory/Optional	Type	Description
`data_type_id`	Mandatory	string	The data type identifier should be unsubscribed for its availability notifications
`rApp_id`	Mandatory	string	The identifier of the registered rApp

Table 28
Functional Outputs

Status	Type	Description
Successful	dictionary	Empty dictionary upon successful operation
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.5
Discover Data Type

Method Name	`discover_dataType(self, discoverDataTypeDetails)`

Description

This method discovers all data type IDs, excluding those registered by caller of the API.

Table 29
Input Parameters

Key	Null	Type	Description
`discoverDataTypeDetails`	n/a	discoverDataType	Data model that contains the filter criteria of the searched data types

Table 30
Functional Outputs

Status	Type	Description
Successful	list[string]	List of data type IDs which are discovered
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.6
Subscribe Data

Method Name	`subscribe_data(self, subscribeDataDetails)`

Description

This method subscribe to data registered with the NonRT-RIC framework.

Table 31
Input Parameters

Key	Mandatory/Optional	Type	Description
`subscribeDataDetails`	Mandatory	subscribeDataRequest	Data model that contains the details for subscribing to service details

Table 32
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing the information about the subscribed data in the following format:

Key	Type	Description
`accessMethod`	string	The type of data access available can be `WS`, `RESTApi`, `MsgBus`
`accessDetails`	dictionary	Access method details contain the following keys: `url`, `topic`, `token`.
`interfaceEndpoint`	string	URL for accessing the producer
`collectionMethod`	string	The type of `collectionMethod` can vary between `Push-Pull`, `Push`, `Pull`
`collectionPeriod`	integer	Planned Future Enhancement
`scope`	string	Planned Future Enhancement
`target`	string	Planned Future Enhancement

Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.7
Pull Data

Method Name	`pullData(self, pullDetails)`

Description

This method pull data from a data type registered with the NonRT-RIC framework.

Table 33
Input Parameters

Key	Mandatory/Optional	Type	Description
`pullDetails`	Mandatory	pullDataRequest	Data model that contains the payload for pull data API

Table 34
Functional Outputs

Status	Type	Description
Successful	list[dictionary]	List of dictionaries containing the fetched data from Laguna, the format may vary based on the data type
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.8
Push Data

Method Name	`pushData(self, pushDetails)`

Description

This method push data to a data type registered with the NonRT-RIC framework.

Table 35
Input Parameters

Key	Mandatory/Optional	Type	Description
`pushDetails`	Mandatory	pushDataRequest	Data model that contains the payload for push data API

Table 36
Functional Outputs

Status	Type	Description
Successful	dictionary	The dictionary contains the timestamp key in the following format: Key: `timestamp` — Type: string — Description: The push timestamp can be later used as Laguna reference in pull data to be able to access the pushed data instance
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.9
Request Data

Method Name	`request_data(self, requestDetails)`

Description

This method request data to a data type registered with the NonRT-RIC framework.

Table 37
Input Parameters

Key	Mandatory/Optional	Type	Description
`requestDetails`	Mandatory	requestDataRequest	Data model that contains the payload for request data API

Table 38
Functional Outputs

Status	Type	Description
Successful	dictionary	Query generated from the input fields given in `requestDetails`
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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4.10
Get All Data Type IDs

Method Name	`getAll_dataType(self, getAllDataTypeDetails)`

Description

This method get a single/all data type in the NonRT-RIC framework.

Table 39
Input Parameters

Key	Mandatory/Optional	Type	Description
`getAllDataTypeDetails`	Mandatory	getAllDataType	Details of the request to get all data types

Table 40
Functional Outputs

Status	Type	Description
Successful	list	List of dictionaries containing all the `dataType` information in the following format: if `showAll` is true: **Key** / **Type** / **Description** `_id` / string / Primary key for data type identifier `dataTypeId` / string / The data type identifier `rAppName` / string / The name of producer rApp `accessMethod` / string / Access method can be either `RESTApi`, `WS`, or `MsgBus` (Kafka, which is in the deprecating path) `collectionMethod` / string / If the `accessMethod` is `RESTApi`, the collection method can be either `Push` or `Pull`. Otherwise, this value does not matter `availibilityStatus` / enum / Placeholder for future expansion. It is always available. `collectionPeriod` / integer / The value 0 is a placeholder for future updates If `showAll` is False, returns a list of `_id`

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Table 40 (Cont.)
Functional Outputs

Status	Type	Description
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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Chapter 5
Configuration Management and Exposure Package

In this chapter:

5.1 Read Configuration
5.2 Write Configuration
5.3 Query Alarm Info
5.4 Query Performance Info

The following methods of the SDK are to support some of the APIs exposed by Configuration Management and Exposure (CME).

Note: To use CME SDK functions, the CME client instance must be created from the CME client class provided by the rApp SDK by adding the following codes in the Python file.

Command:
```
from rappsdk.cme.cme_client import CMEClient
cme_client_instance = CMEClient(cme_url, username, password)
```

Table 41
Input Parameters

Key	Mandatory/Optional	Type	Description
`cme_url`	Mandatory	string	CME server URL
`username`	Mandatory	string	The username is registered in keycloak
`password`	Mandatory	string	Password

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5.1
Read Configuration

Method Name	`read_configuration(self, readConfigurationDetails)`

Description

This method allows rApp to read the configuration of any network device through Virtuora® Mobility Controller (MC) and return its configurations.

This method takes a dictionary as an input, a sample input can be found in the following table.

Table 42
Input Parameters

Key	Mandatory/Optional	Type	Description
`readConfigurationDetails`	Mandatory	readConfigurationDetails	Dictionary containing the configuration details needed for reading with CME

Table 43
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing configurations of the given device
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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5.2
Write Configuration

Method Name	`write_configuration(self, writeConfigurationDetails)`

Description

This method allows rApp to modify the configuration of any network devices through Virtuora MC.

This method takes a dictionary as an input.

Table 44
Input Parameters

Key	Mandatory/Optional	Type	Description
`writeConfigurationDetails`	Mandatory	writeConfigurationDetails	Dictionary containing the configuration details needed for writing with CME.

Table 45
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing configurations of the given device
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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5.3
Query Alarm Info

Method Name	`query_performance_info(self, queryAlarmInfoDetails)`

Description

This method allows rApp to fetch the current alarm data of the given network device from the Laguna datastore.

This method takes a dictionary as an input, a sample input can be found in the following table.

Table 46
Input Parameters

Key	Mandatory/Optional	Type	Description
`queryAlarmInfoDetails`	Mandatory	queryAlarmInfoDetails	Dictionary containing the query details needed for fetching the information from the CME

Table 47
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing current performance data of the given network device
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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5.4
Query Performance Info

Method Name	`query_performance_info(self, queryPerformanceInfoDetails)`

Description

This method allows rApp to fetch the current performance data of the given network device from the Laguna datastore.

This method takes a dictionary as an input, a sample input can be found in the following table.

Table 48
Input Parameters

Key	Mandatory/Optional	Type	Description
`queryPerformanceInfoDetails`	Mandatory	queryPerformanceInfoDetails	Dictionary containing the query details needed for fetching the information from the CME

Table 49
Functional Outputs

Status	Type	Description
Successful	dictionary	Dictionary containing current performance data of the given network device
Unsuccessful	Exceptions	Method raises an instance of exception if the operation is unsuccessful

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Chapter 6
Schema

In this chapter:

6.1 rAppRegister
6.2 serviceTypeEnum
6.3 serviceRegister
6.4 accessMethodsEnum
6.5 serviceUpdate
6.6 subscribeRequest
6.7 unSubscribeService
6.8 dataProducerLimits
6.9 accessMethodEnum
6.10 pushPull
6.11 status
6.12 transferMethod
6.13 subscribeDataTypeRequest
6.14 discoverDataType
6.15 dataTypeRegister
6.16 subscribeDataRequest
6.17 pullDataRequest
6.18 pushDataRequest
6.19 requestDataRequest
6.20 getAllDataType
6.21 readConfigurationDetails
6.22 writeConfigurationDetails
6.23 queryAlarmInfoDetails
6.24 queryPerformanceInfoDetails

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6.1
rAppRegister

Class Name	`rAppRegister`
Full Path	*`rappsdk.sme_schemas.rapp.rAppRegister`*

Description

The data model for rApp registration information.

Table 50
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`rAppName`	Mandatory	string	—	Name of the rApp
`vendor`	Mandatory	string	—	Name of vendor
`rAppVersion`	Mandatory	string	—	Version of rApp
`certificates`	Optional	dictionary	—	Certificates. Currently this can be none
`serviceType`	Mandatory	List[serviceTypeEnum]	`"serviceConsumer"`, `"serviceProducer"`	List of instances of `serviceTypeEnum`
`appType`	Mandatory	string	—	Type of application, for example, MLAnalysis
`interfaceEndpoint`	Mandatory	string	—	URL endpoint of rApp where it is deployed

Note: Instance or object of this class is input to Register rApp.

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6.2
serviceTypeEnum

Class Name	`serviceTypeEnum`
Full Path	*`rappsdk.sme_schemas.enums.serviceTypeEnum`*

Description

`serviceTypeEnum` is an Enumeration class to indicate the type of service that rApp provides.

Table 51
Member

Value	Description
`serviceConsumer`	Represents rApp of the type service consumer
`serviceProducer`	Represents rApp of the type service producer

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6.3
serviceRegister

Class Name	`serviceRegister`
Full Path	*`rappsdk.sme_schemas.service.serviceRegister`*

Description

The data model for service registration information.

Table 52
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Id of the rApp
`serviceName`	Mandatory	string	Name of the service
`serviceVersion`	Mandatory	string	Version of the service
`accessMethod`	Mandatory	accessMethodsEnum	`accessMethod` for the given service
`communicationMechanism`	Mandatory	dictionary	Communication mechanism. Currently, dictionary is empty
`serviceEndpoint`	Mandatory	string	URL or endpoint of the service
`serviceLimitations`	Mandatory	dictionary	Limitations of the service. Currently, dictionary is empty
`serviceCapabilities`	Mandatory	dictionary	Capabilities of the service. Currently, dictionary is empty

Note: Instance or object of this class is input to Service Registration.

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6.4
accessMethodsEnum

Class Name	`accessMethodsEnum`
Full Path	*`rappsdk.sme_schemas.enums.accessMethodsEnum`*

Description

`accessMethodsEnum` is an Enumeration class to indicate the access types for the services and data types.

Table 53
Member

Value	Description
`MsgBus`	Indicates that message bus is used for communication
`RESTApi`	Indicates that RestAPI is used for communication

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6.5
serviceUpdate

Class Name	`serviceUpdate`
Full Path	*`rappsdk.sme_schemas.service.serviceUpdate`*

Description

The data model for updating service details.

Table 54
Input Parameters

Key	Mandatory/Optional	Type	Description
`serviceVersion`	Optional	string	Version of the service
`accessMethod`	Optional	accessMethodsEnum	Indicates the access type for the given service
`communicationMechanism`	Optional	dictionary	Indicates the communication mechanism for the given service/data type
`serviceEndpoint`	Optional	string	URL of endpoint of the service
`serviceLimitations`	Optional	dictionary	Limitations of the service
`serviceCapabilities`	Optional	dictionary	Capabilities of the service

Note: Instance or object of this class is input to Service Update.

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6.6
subscribeRequest

Class Name	`subscribeRequest`
Full Path	*`rappsdk.sme_schemas.subscribe.subscribeRequest`*

Description

The data model for subscribing service information.

Table 55
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Name of the rApp
`subscribedServices`	Mandatory	list[string]	List of the service names which have to be subscribed

Note: Instance or object of this class is input to Subscribe Service.

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6.7
unSubscribeService

Class Name	`unSubscribeService`
Full Path	*`rappsdk.sme_schemas.subscribe.unSubscribeService`*

Description

The data model for unsubscribe service information.

Table 56
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	`rAppId` which has to unsubscribe from the service
`subscribedServices`	Mandatory	list[string]	List of the services which have been subscribed
`subscriptionId`	Mandatory	string	`subscriptionId` of the service which must be unsubscribed

Note: Instance or object of this class is input to Unsubscribe Service.

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6.8
dataProducerLimits

Class Name	`dataProducerLimits`
Full Path	*`rappsdk.dme_schemas.data_type.dataProducerLimits`*

Description

The data model to set limits for data producer.

Table 57
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`maxPushTime`	Mandatory	integer	`1…2147483647`	It can be set to 0 if the value is not known
`maxConsumers`	Mandatory	integer	`1…2147483647`	The number of maximum consumers rApp can serve
`maxTransferRate`	Mandatory	integer	`1…2147483647`	It can be set to 0 if the value is not known

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6.9
accessMethodEnum

Class Name	`accessMethodsEnum`
Full Path	*`rappsdk.dme_schemas.enums.accessMethodsEnum`*

Description

`accessMethodsEnum` is an Enumeration class to indicate the access types for the services and data types.

Table 58
Member

Value	Description
`MsgBus`	Indicates that message bus is used for communication
`RESTApi`	Indicates that RestAPI is used for communication

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6.10
pushPull

Class Name	`pushPull`
Full Path	*`rappsdk.dme_schemas.enums.pushPull`*

Description

`pushPull` is the Enumeration class to indicate the collection method type for the data transfer.

Table 59
Member

Value	Description
`pull`	Pull is the producer required collection method. It means the new data instance is provided to DME using producer `pull_data()` API
`push`	Push is the producer required collection method. It means the new data instance (corresponding to the data type) can be sent directly to DME using DME `push_data()` API

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6.11
status

Class Name	`status`
Full Path	*`rappsdk.dme_schemas.enums.status`*

Description

`status` is an Enumeration class to indicate the rApp status.

Table 60
Member

Value	Description
`active`	Corresponds to value `ACTIVE`, which indicated that rApp is active
`inactive`	Corresponds to value `INACTIVE`, which indicated that rApp is inactive

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6.12
transferMethod

Class Name	`transferMethod`
Full Path	*`rappsdk.dme_schemas.enums.transferMethod`*

Description

`transferMethod` is an Enumeration class to indicate the transfer method.

Table 61
Member

Value	Description
`transferMethodA`	Corresponds to value `TRANSFER_METHOD_A`
`transferMethodB`	Corresponds to value `TRANSFER_METHOD_B`

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6.13
subscribeDataTypeRequest

Class Name	`subscribeDataTypeRequest`
Full Path	*`rappsdk.dme_schemas.data_type.subscribeDataTypeRequest`*

Description

The data model to subscribe data type.

Table 62
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Unique identifier of rApp, which is generated and returned by SME when registering rApp
`interfaceEndpoint`	Mandatory	string	Input must be in valid URL format, for example, *`http://rapp.<ip address>.nip.io/`*. An external rApp or microservice communicates with this rApp through this URL (API communication). It can be empty string if the value is not known
`dataTypeId`	Mandatory	string	`dataTypeId` can be any unique alphanumeric string. It cannot be an empty string.
`collectionMethod`	Mandatory	string or pushPull	Method to be used by rApp
`scope`	Mandatory	string	—
`target`	Mandatory	string	—

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6.14
discoverDataType

Class Name	`discoverDataType`
Full Path	*`rappsdk.dme_schemas.data_type.discoverDataType`*

Description

The data model to discover data type.

Table 63
Input Parameters

Key	Mandatory/Optional	Type	Description
`rAppId`	Mandatory	string	Unique identifier of rApp, which is generated and returned by SME when registering rApp
`producerLimitsMaxPushTime`	Optional	integer	—
`producerLimitsMaxConsumers`	Optional	integer	—
`dataSchema`	Optional	string	The `dataTypeId` can be any unique alphanumeric string. It cannot be an empty string
`collectionMethod`	Optional	string or pushPull	Method to be used by rApp

Note: Instance or object of this class is input to Discover Data Type.

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6.15
dataTypeRegister

Class Name	`dataTypeRegister`
Full Path	*`rappsdk.dme_schemas.data_type.dataTypeRegister`*

Description

The data model to discover data type.

Table 64
Input Parameters

Key	Mandatory /Optional	Type	Value	Description
`dataTypeId`	Mandatory	string	—	`dataTypeId` can be any unique alphanumeric string. It cannot be an empty string.
`rAppId`	Mandatory	string	—	Unique identifier of rApp which is generated and returned by SME when registering rApp.
`accessMethod`	Mandatory	string	`"RESTApi",` `"MsgBus", "WS"`	DME supports three access methods. String should be one of those three.
`collectionMethod`	Mandatory	string	`"PUSH", "PULL"`	PUSH: Push is the producer required collection method. It means the new data instance (corresponding to the data type) can be sent directly to DME using DME `push_data()` API. PULL: Pull is the producer required collection method. It means the new data instance is provided to DME using producer `pull_data()` API.
`availibilityStatus`	Mandatory	string	`"ACTIVE" ,` `"INACTIVE"`	Activity status of the data type
`collectionPeriod`	Mandatory	integer	`1… 2147483647`	Input must be an integer. It can be set to 0 if the value is not known.
`dataSchema`	Mandatory	list[dictionary]	—	Schema of the data type. Data corresponding to the data type should always respect the given schema.

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Table 64 (Cont.)
Input Parameters

Key	Mandatory /Optional	Type	Value	Description
`producerLimits`	Mandatory	dictionary	—	Dictionary contains information related to serving the limitations of producers with the following structure. Note that all these fields are mandatory. If the value is not known, it can be set to 0. **Key** / **Type** / **Value** / **Description** `maxPushTime` / integer / `1…2147483647` / Can be set to 0 if the value is not known `maxConsumers` / integer / `1…2147483647` / Number of maximum consumers rApp can serve `maxTransferRate` / integer / `1…2147483647` / Can be set to 0 if the value is not known

`producerCapabilities`	Mandatory	string	`"TRANSFER_METHOD_A",`
			
			`"TRANSFER_METHOD_B"`
				Input can be either `TRANSFER_METHOD_A` or `TRANSFER_METHOD_B` (PEE)

Note: Instance or object of this class is input to Register Data Type.

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6.16
subscribeDataRequest

Class Name	`subscribeDataRequest`
Full Path	*`rappsdk.dme_schemas.subscribe_data.subscribeDataRequest`*

Description

A class of schemas for DME when subscribing to a data type.

Table 65
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`rAppId`	Mandatory	string	—	Input must be the `rAppId` of the rApp which would be like to subscribe for the data type
`interfaceEndpoint`	Mandatory	string	—	Input must be in valid URL format, for example, *`http://rapp.<ip address>.nip.io/`*. An external rApp or microservice communicates with this rApp through this URL (API communication). It can be empty string if the value is not known
`dataTypeId`	Mandatory	string	—	Id of the data type to which the rApp would be like to subscribe
`collectionMethod`	Mandatory	string	—	Input can be either `PUSH` or `PULL`
`collectionPeriod`	Mandatory	integer	`1… 2147483647`	Input must be an integer. It can be set to 0 if the value is not known
`scope`	Mandatory	string	—	Input must be a string. The empty string can be sent as input if the value is not known
`target`	Mandatory	string	—	Input must be a string. The empty string can be sent as input if the value is not known

Note: Instance or object of this class is input to Subscribe Data Type Change.

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6.17
pullDataRequest

Class Name	`pullDataRequest`
Full Path	*`rappsdk.dme_schemas.pull_data.pullDataRequest`*

Description

A class of schemas for DME to pull data.

Table 66
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataTypeId`	Mandatory	string	Input must be referred to pull its corresponding data instance
`laguna_ref`	Mandatory	string/dictionary	Reference to the data type instance. It might refer to a single data point or bulk data/query in dictionary format generated by Request Data

Note: Instance or object of this class is input to Pull Data.

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6.18
pushDataRequest

Class Name	`pushDataRequest`
Full Path	*`rappsdk.dme_schemas.push_data.pushDataRequest`*

Description

A class of schemas for DME to push data.

Table 67
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataTypeId`	Mandatory	string	Identifier of the targeted data type
`data`	Mandatory	string	The actual single data instance should be matching with the original schema of the data type

Note: Instance or object of this class is input to Push Data.

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6.19
requestDataRequest

Class Name	`requestDataRequest`
Full Path	*`rappsdk.dme_schemas.request_data.requestDataRequest`*

Description

A class of schemas for DME to request data.

Table 68
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`dataTypeId`	Mandatory	string	—	Laguna store name or RestAPI-based `dataTypeId` name
`startTime`	Mandatory	datetime/dictionary	—	`startTime` can take a datetime input or dictionary. If the dictionary is given, it selects the column for timestamp based on the key provided for the dictionary; otherwise, it takes the default timestamp column `timestampnew`
`stopTime`	Mandatory	datetime	—	`stopTime` can take a datetime input or dictionary. If the dictionary is given, it selects the column for timestamp based on the key provided for the dictionary; otherwise, it takes the default timestamp column `timestampnew`
`timeInterval`	Mandatory	integer	`1… 2147483647`	Planned Future Enhancement
`period`	Mandatory	integer	`1… 2147483647`	Planned Future Enhancement
`target`	Mandatory	dictionary	—	The target should be provided as a dictionary. The dictionary key is selected as the primary index for filtering
`scope`	Mandatory	list[dictionary]	—	The scope is a list of dictionaries. Each dictionary must have the column name as the key that is used as the secondary index for filtering, followed by the value that must be filtered based on

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Table 68 (Cont.)
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`result`	Mandatory	list	—	The result is a list that takes the names of the columns that must be the only values in the final data pull

Note: Instance or object of this class is input to Request Data.

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6.20
getAllDataType

Class Name	`getAllDataType`
Full Path	*`rappsdk.dme_schemas.data_type.getAllDataType`*

Description

A class of schemas for DME to get data type.

Table 69
Input Parameters

Key	Mandatory/Optional	Type	Value	Description
`rAppId`	Optional	string	—	Optional parameter if include all of the available data type IDs are filtered based on the corresponding given `rAppID`. If not given, all of the available data type IDs are returned.
`showAll`	Optional	boolean	`False`	(default) The returned list only includes the data type IDs
			`True`	More information corresponding to each data type identifier is included

Note: Instance or object of this class is input to Get All Data Type IDs.

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6.21
readConfigurationDetails

Description

Input JSON to read the configuration of any network devices through Virtuora MC.

Table 70
Input Parameters

Key	Mandatory/Optional	Type	Description
`ManagedElementId`	Optional	string	Id of the managed element/network device whose configuration must be fetched
`GNBDUFunctionId`	Optional	string	A unique identifier identify Next Generation Node B (gNB) Distributed Unit (DU) uniquely within gNB Central Unit (CU)
`NRCellDUId`	Optional	string	Id of the cell attached to the given DU whose configurations have to be fetched

Note: Instance or object of this class is input to Read Configuration.

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6.22
writeConfigurationDetails

Description

Input JSON to update the configuration of any network devices through Virtuora MC.

Table 71
Input Parameters

Key	Mandatory/Optional	Type	Description
`ManagedElementId`	Mandatory	string	Id of the managed element/network device whose configuration must be fetched
`GNBDUFunctionId`	Mandatory	string	A unique identifier identify gNB DU uniquely within gNB CU
`NRCellDUId`	Mandatory	string	Id of the cell attached to the given DU whose configurations have to be fetched
`ManagedNFService`	Optional	string	Id of `ManagedNFService` under `NRCellDU`, whose configuration must be updated
`data`	Mandatory	dictionary	JSON containing the id and configuration parameter attributes to be updated within the managed network function list

Note: Instance or object of this class is input to Write Configuration.

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6.23
queryAlarmInfoDetails

Description

Input JSON to fetch the current alarm data of the given network device from the Laguna datastore.

Table 72
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataType`	Mandatory	string	`dataTypeId` of the data to be fetched. The `dataTypeId` can be any alphanumeric string. It cannot be an empty string.
`target`	Mandatory	dictionary	Accepts a dictionary whose fields have to be searched for in the database, and only the data which has this key-value pair in it is returned. Acts as the primary filter for the data
`scope`	Mandatory	list[dictionary]	Accepts a list of dictionaries indicating a secondary filter for the data. All the data having any of these dictionaries mentioned in the scope is returned
`result`	Mandatory	list[string]	Accepts the parameters/fields from the data which must be returned
`timeIndex`	Mandatory	string	Index for the time field on which the time range filter must be applied

Note: Instance or object of this class is input to Query Alarm Info.

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6.24
queryPerformanceInfoDetails

Description

Input JSON to fetch the current performance data of the given network device from the Laguna datastore.

Table 73
Input Parameters

Key	Mandatory/Optional	Type	Description
`dataType`	Mandatory	string	`dataTypeId` of the data to be fetched. The `dataTypeId` can be any alphanumeric string. It cannot be an empty string
`target`	Mandatory	dictionary	Accepts a dictionary whose fields have to be searched for in the database, and only the data which has this key-value pair in it is returned. Acts as the primary filter for the data
`scope`	Mandatory	list[dictionary]	Accepts a list of dictionaries indicating a secondary filter for the data. All the data having any of these dictionaries mentioned in the scope is returned
`result`	Mandatory	list[string]	Accepts the parameters/fields from the data which must be returned
`timeIndex`	Mandatory	string	Index for the time field on which the time range filter must be applied

Note: Instance or object of this class is input to Query Performance Info.

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Chapter 7
Exceptions

This section describes the exceptions that could occur from SME, DME, or CME API calls that are invoked by the SDK functions.

Table 74
Exceptions

Errors	Description
Authentication	Authentication error is raised if authorization fails on the server side when invoking the API.
Authorization	Authorization error is raised if the given user credentials are not authorized for accessing the invoked API.
Internal Server	Internal server error is raised if the server is down or is not able to process the request.
Validation	Validation error is raised if the server is not able to process the request because of semantic errors in request
Duplicate	Duplicate error is raised if the request was already made and the user is trying to send the same request again

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Appendix A
Glossary
Terms and Definitions

This chapter describes the product glossary terms and definitions.

API	Application Programming Interface
CME	Configuration Management and Exposure
CU	Central Unit
DME	Data Management and Exposure
DU	Distributed Unit
rApp	NonRT-RIC Application
SDK	Software Development Kit
SME	Service Management and Exposure

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